

SHELDON, IA, USA

WMO: 725495

Lat: 43.208N

Lon: 95.833W

Elev: 433

StdP: 96.24

Time Zone: -6.00 (NAC)

Period: 95-19

WBAN: 04975

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.0	-20.9	-27.3	0.3	-23.1	-24.1	0.4	-20.4	14.9	-6.1	13.6	-8.0	3.1	330	0.610

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.8	32.7	24.1	31.3	23.4	30.0	22.9	26.6	30.3	25.4	29.3	24.3	28.3	5.9	190

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.3	21.5	28.8	24.0	19.9	27.9	22.9	18.5	26.9	86.2	30.8	80.3	29.6	76.2	27.5	30.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 12.7	11.4	10.2	-29.5	35.4	2.9	2.0	-31.6	36.8	-33.3	38.0	-34.9	39.1	-37.0	40.6		
(5)			-28.8	28.0	3.0	1.1	-31.0	28.7	-32.7	29.4	-34.4	30.0	-36.6	30.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.4	-7.5	-5.7	1.3	8.7	15.4	21.1	22.7	20.8	16.9	9.8	1.8	-4.9		
	DBStd	12.04	7.00	7.17	7.05	5.70	5.23	3.82	3.29	3.19	4.73	5.45	6.13	6.47		
	HDD10.0	2153	542	438	280	90	12	0	0	0	5	71	252	462		
	HDD18.3	4019	801	672	528	293	123	16	5	12	82	270	497	720		
	CDD10.0	1583	0	0	10	49	178	334	395	335	211	64	6	0		
	CDD18.3	407	0	0	0	3	31	101	142	89	38	4	0	0		
	CDH23.3	4171	0	0	6	62	387	1071	1434	774	373	62	2	0		
	CDH26.7	1404	0	0	1	18	141	406	515	213	97	13	0	0		
(14) Wind	WSAvg	4.5	4.9	5.0	5.3	5.8	5.5	4.5	3.2	2.9	3.5	4.3	4.7	4.8		
(15) Precipitation	PrecAvg	760	17	17	43	78	91	116	102	82	78	54	34	21		
	PrecMax	1062	36	40	92	147	147	229	247	185	246	139	99	54		
	PrecMin	599	0	0	0	20	32	32	8	14	4	4	0	0		
	PrecStd	112	12	11	26	32	36	51	54	49	53	40	29	16		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.8	14.0	23.7	28.8	33.8	34.9	34.1	32.6	32.2	28.7	22.2	12.5		
		MCWB	5.6	10.3	13.9	17.8	19.9	23.6	26.5	26.5	21.9	17.7	14.0	9.6		
	2%	DB	5.0	9.8	18.7	24.8	30.2	32.6	32.5	30.9	29.1	25.0	17.5	8.1		
		MCWB	2.9	7.7	12.0	14.9	18.9	22.9	26.0	25.0	21.4	16.0	11.5	5.2		
	5%	DB	2.7	6.0	15.0	22.0	27.5	31.1	31.2	28.9	27.5	22.1	13.8	5.1		
		MCWB	1.6	3.9	10.4	13.2	17.7	21.8	25.2	24.4	19.8	14.6	8.9	3.0		
	10%	DB	1.3	3.0	11.3	18.6	24.9	29.0	29.8	27.5	25.3	18.9	11.0	2.8		
		MCWB	0.5	1.8	8.3	11.8	16.5	20.8	24.4	22.7	18.6	13.9	7.3	1.5		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.0	11.4	16.7	18.7	23.0	26.6	29.1	27.8	24.6	20.8	15.5	10.1		
		MCDB	8.7	13.5	20.7	27.6	28.8	31.0	32.0	30.6	28.6	24.2	20.3	11.9		
	2%	WB	3.3	7.1	13.5	16.0	20.8	25.0	27.2	26.0	23.2	17.9	12.4	5.6		
		MCDB	4.9	9.4	17.2	22.4	26.8	30.1	30.4	29.4	27.0	21.5	16.8	7.9		
	5%	WB	1.9	4.0	11.0	14.2	19.4	23.8	25.9	24.8	21.9	16.0	9.7	3.3		
		MCDB	3.0	5.8	14.9	19.7	25.3	28.8	29.8	28.0	25.8	20.3	13.3	5.1		
	10%	WB	0.6	2.2	8.0	12.7	17.9	22.5	24.9	23.5	20.5	14.0	7.4	1.7		
		MCDB	1.3	3.4	11.1	17.4	22.8	27.1	28.8	26.6	23.5	18.2	10.3	3.1		
(35) Mean Daily Temperature Range	MDBR		9.3	9.4	10.2	12.3	12.3	12.0	11.8	11.5	13.2	12.3	10.9	9.0		
	5% DB	MCDBR	11.0	11.2	16.2	17.7	16.3	13.9	12.4	12.0	14.7	16.5	16.0	12.1		
		MCWBR	9.5	8.7	10.3	9.4	7.3	6.2	7.1	7.4	7.7	8.8	10.1	9.5		
	5% WB	MCDBR	10.4	10.3	15.1	14.4	13.2	11.8	11.0	10.7	11.7	13.2	14.7	11.2		
(40) Clear-Sky Solar Irradiance	taub		0.268	0.282	0.316	0.365	0.388	0.399	0.409	0.406	0.361	0.328	0.291	0.274		
	taud		2.440	2.421	2.418	2.306	2.316	2.346	2.306	2.293	2.396	2.467	2.493	2.449		
	Ebn at Noon		881	925	928	900	885	873	859	849	867	857	844	837		
	Edh at Noon		73	89	102	123	126	123	126	124	103	85	70	67		
(44) All-Sky Solar Radiation	RadAvg		1.90	2.98	3.85	4.82	5.59	6.24	6.61	5.49	4.41	3.00	2.06	1.53		
	RadStd		0.18	0.25	0.32	0.42	0.40	0.44	0.28	0.40	0.30	0.37	0.13	0.14		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (44 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page